

- 9 (a) Fig. 9.1 below shows a Rutherford scattering experiment in which α -particles are directed at a gold foil. The detector is shown in two positions in the evacuated chamber.

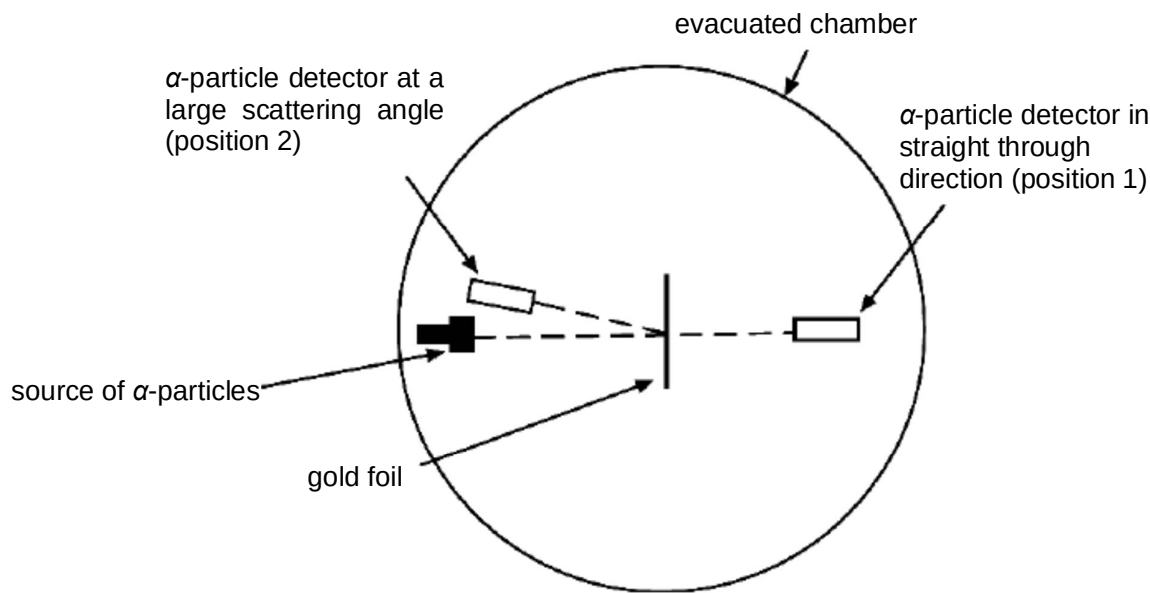


Fig. 9.1

- (i) Explain why air needs to be removed from the apparatus.

to minimise collisions between α particles and air molecules) B1

Or

α particles have a short range in air (3–5 cm) B1

(or any answer to the same effect)

- (ii) Explain why the gold foil should be very thin.

the α particles must only be scattered at most once B1

Or

- the α particles must not be absorbed/stopped/blocked by the foil B1

("to allow the alpha particle to emerge from the gold foil" is not accepted)

- (iii) 1. Explain why a high count rate is detected by the α -particle detector in position 1.

Most of the α particles pass straight through as the atom consists mainly of open space B1

(do not accept small nucleus)

2. A low count rate is detected by the α -particle detector in position 2. State what can be deduced from this observation about the structure of the atom and the properties of the gold nucleus.

- the nucleus is positively charged

- the nucleus is **very small in comparison to the size of an atom**

- the nucleus is very massive

(any two answers)

- (b) Fig. 9.2 illustrates the variation with nucleon number A of the binding energy per nucleon E of nuclei.

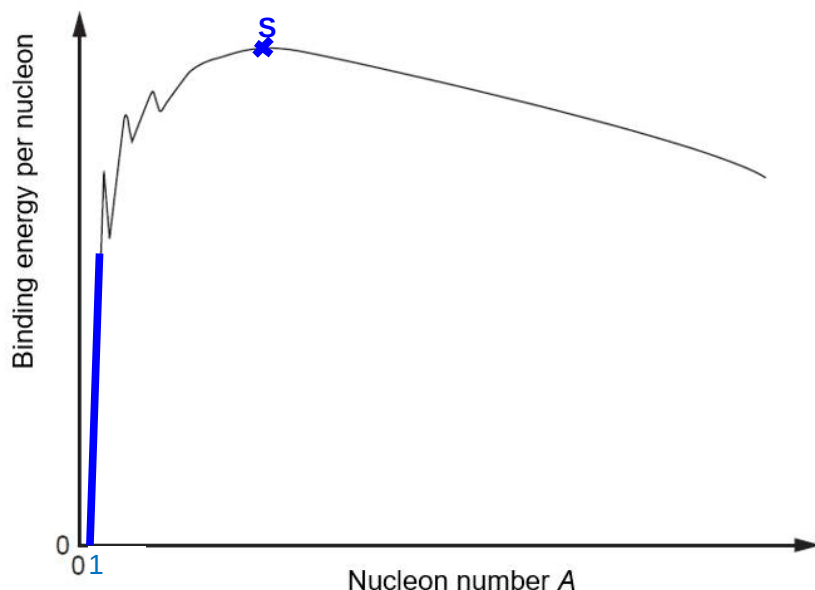


Fig. 9.2

- (i) Explain what is meant by the *binding energy* of a nucleus.

Binding energy of a nucleus refers to the amount of energy required to break it apart into its separate constituents of protons and neutrons. **B1**

nucleons separated to infinity / completely **B1**

- (ii) With the aid of Fig. 9.2, explain why more energy per nucleon is released in fusion than in fission.

- Energy released in a nuclear reaction is equal to the difference in binding energies between the products and the original reactants. **B1**

- Fusion occurs for reactants and products that have nucleon numbers less than 56 while fission occurs for reactants and products that have nucleon numbers greater than 56. **B1**

Or

- Fusion reaction is one in which two light nuclei combine to form a heavier nucleus, while in fission a heavy nucleus splits into two lighter nuclei. **B1**

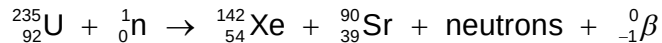
- From graph, the steeper slope of the binding energy curve for lighter nuclei indicates that the change in binding energy in fusion is larger, compared to that for fission reactions. **B1**

- Hence, more energy is released in a fusion reaction.

- (iii) On Fig. 9.2, mark with the letter **S** the region of the graph representing nuclei having the greatest stability. [1]

- (iv) On Fig. 9.2, complete the graph for elements of low nucleon numbers. [1]

- (c) Uranium-235 may undergo fission when bombarded by a slow moving neutron to produce Xenon-142 and Strontium-90 as shown below.



- (i) Determine the number of neutrons produced in this fission reaction.

No of neutrons = 4 **A1**

- (ii) State the role of these neutrons in the fission reaction.

The role of neutrons is to initiate more nuclear fission reactions by bombarding other ${}_{92}^{236}\text{U}$ nuclei such that a chain reaction results.

- (iii) Data on binding energy per nucleon for some nuclides are given in Fig. 9.3.

isotope	binding energy per nucleon / MeV
Uranium-235	7.59
Xenon-142	8.23
Strontium-90	8.70

Fig. 9.3

Calculate

1. the energy released in this fission reaction,

$$\begin{aligned} \text{energy released} &= (8.23 \times 142 + 8.70 \times 90) - 235 \times 7.59 \dots\dots\dots \mathbf{C1} \\ &= 168 \text{ MeV} \\ &= 2.69 \times 10^{-11} \text{ J} \dots\dots\dots \mathbf{A1} \end{aligned}$$

2. the mass equivalent of this energy.

$$\begin{aligned} \text{mass equivalent} &= (2.69 \times 10^{-11}) / (3.0 \times 10^8)^2 \\ &= 2.99 \times 10^{-28} \text{ kg} \dots\dots\dots \mathbf{A1} \end{aligned}$$

- (iv) A nuclear power station supplies 100 MW of electrical power to a town for one year at an efficiency of 35%.

Calculate the mass of uranium needed to operate the power station for one year.

Useful output Power = $100 \times 10^6 \text{ W}$ at 35% efficiency

In a year, total energy required = $100 \times 10^6 \times 365 \times 24 \times 60 \times 60 / 0.35$

$$= 9.01 \times 10^{15} \text{ J} \dots\dots\dots \mathbf{C1}$$

Energy released in 1 reaction = $2.69 \times 10^{-11} \text{ J}$

No of reactions required = $9.01 \times 10^{15} / 2.69 \times 10^{-11}$

= 3.35×10^{26} **C1**

= N, number of uranium atoms required.

Mass required = $3.35 \times 10^{26} \times 235 \times 1.66 \times 10^{-27}$

= $1.3 \times 10^2 \text{ kg}$ **A1**